

Does a Music Transformer Know What Key It Is In?

Causal Interventions on an Emergent Tonal World Model

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Abstract—Music generation models produce convincingly tonal music, but it is unknown whether they merely chain surface statistics or maintain an internal representation of musical key that is *causally used* during generation. Following the Othello-GPT methodology, we train a 25M-parameter decoder-only Transformer on symbolic music from a functional-harmony grammar whose vocabulary contains absolute pitch and time only — no key, chord, or degree tokens — and test the world-model hypothesis at three levels, under a pre-registered protocol with frozen decision rules. (i) *Existence*: linear probes decode the running key from mid-layer activations at 0.93 macro- F_1 , exceeding the strongest input-side baseline (0.79) in all six trained models, with the margin widening exactly where surface statistics are least informative. (ii) *Geometry*: the twelve tonic directions self-organize into a circle of fifths (neighbor agreement 1.0 vs. 0.0 for the chromatic circle); transposition acts as a generalizing linear map, yet these maps do not compose cyclically, and transposition augmentation does not change this — a pre-registered negative result. (iii) *Causality*: replacing the key component of the residual stream at a single middle layer makes the model continue in the injected key at $5.6\times$ the rate of rank- and norm-matched random edits (0.38 vs. 0.07, significant for 12/12 target keys after Holm correction) while staying within a frozen musicality budget — about 58% of the behavioral upper bound obtained by transposing the prompt itself. A capacity sweep sharpens the claim: next-token accuracy is flat from 1.6M to 85M parameters, yet below ~ 6 M the key state does not exist — behaviorally equivalent models can be pure surface statistics, and the pre-registered test tells them apart. Editing a music Transformer’s internal key state changes what key it composes in: tonality is tracked as a causal latent variable, not just surface statistics.

Index Terms—music generation, world models, interpretability, causal intervention, tonality

I. INTRODUCTION

Sequence models trained only on next-token prediction can build internal models of the latent state of their domain: Othello-GPT tracks the board implied by move sequences, and editing that internal board changes the moves the model considers legal [1], [2]. Analogous evidence exists for chess, mazes, and RL environments [3], [4], [5], [6], [7], but not for music, where probing studies report *readability* of musical attributes [8] without testing whether the model *uses* the probed state. Music is

TABLE I
TERMS USED THROUGHOUT (CF. SEC. II-A).

key state $\kappa(t)$	tonic ($\mathbb{Z}_{12}$) $\times$ mode (maj/min) at token t
D-SYN	synthetic corpus with per-token ground-truth $\kappa(t)$
M-CTRL / M-REF	studied model / frozen reference for quality
subspace edit	$h \leftarrow h - P_V h + P_V \mu_{\kappa^*}$ in the residual stream
donor key κ^*	key injected by the edit
TKR	rate at which the continuation’s estimated key = κ^*
IKR _{x}	fraction of continuation pitches diatonic to key x
guard δ_{PPL}	M-REF perplexity budget frozen pre-intervention

arguably an ideal substrate for this question: the candidate state variable — the key $\kappa = (\text{tonic}, \text{mode})$ — is defined externally by centuries of theory and by psychological measurement [9], ground-truth state labels can be generated exactly from a functional-harmony grammar, and diatonic scale membership provides a natural analogue of Othello’s legal-move rate.

We ask: *does a music Transformer maintain a causal internal representation of key?* Concretely, we (1) train GPT-2–style models on synthetic tonal music whose token vocabulary provably leaks no key information; (2) probe every layer for the running key against strong input-side baselines and selectivity controls; (3) characterize the geometry of the key representation under the group action of transposition; and (4) causally intervene: we replace the key subspace of the residual stream during generation and measure whether the continuation follows the injected key, against matched random-subspace controls and under a reference-model perplexity budget that disqualifies quality-destroying edits.

The study was pre-registered: all decision rules, thresholds, and controls were frozen before any experiment ran, deviations are logged with reasons, and negative results are reported. Code, data generator, and the full evaluation protocol will be released as a benchmark.

II. METHOD

A. Terminology

Table I fixes vocabulary. Full protocol, decision rules, and all controls are specified in the pre-registration (supplementary).

B. Data and models

D-SYN. A functional-harmony state machine ($T \rightarrow S \rightarrow D \rightarrow T$ with substitutions, SATB voicing, optional diatonic melody) generates 200k/10k/10k train/val/test pieces of 278–508 tokens with 0–3 modulations each (pivot-chord, direct, or sequential; fifths-distance stratified), giving every token a ground-truth key label. **Tokenizer.** The vocabulary (124 types) contains only **BAR**, **POS**, absolute-MIDI **PITCH**, and **DUR** tokens; a unit test enforces the absence of key/chord/degree tokens. **Models.** M-CTRL is a GPT-2 style decoder (8 layers, 8 heads, $d=512$, context 512, ~ 25 M parameters), trained 12k steps under two regimes — with (R-Aug) and without (R-NoAug) uniform 12-fold transposition augmentation — $\times 3$ seeds. M-REF shares the architecture but a disjoint seed and data split and is used exclusively for the musicality guard. All models pass a pre-frozen quality gate (val. next-token top-1 0.878–0.881 vs. a 0.434 constant-predictor rate; unconditional generation in-key ratio 0.964–0.977 vs. a 0.608 threshold).

C. Phase A: probing and geometry

Linear (multinomial logistic) probes map residual-stream activations $h_\ell(t)$ to $\kappa(t)$ (24 classes), with sequence-level splits and three control families: **C1** selectivity (per-sequence permutation of key identities — a control task in the sense of [10]; the literal within-sequence shuffle is also reported), **C2** an untrained model, and **C3** input-side baselines at the same positions (pitch-class-histogram logistic regression and Krumhansl–Schmuckler (KS) matching, windows $W \in \{8, 16, 32, 64\}$). Decision rule DR-H1: the probe supports H1 iff its selectivity-corrected F_1 exceeds the best C3 baseline with a sequence-level BCa 95% CI excluding zero in some layer. Ambiguity is stratified by KS posterior entropy ($W=16$). For geometry, orthogonal Procrustes maps R_k are fit between activations of pieces and their k -semitone transpositions; cyclicity error $\varepsilon_{\text{cyc}} = \text{mean}_k \|R_k - R_1^k\|_F / \|R_k\|_F$ tests group structure (DR-H2b: $\varepsilon_{\text{cyc}}(\text{R-Aug}) < (\text{R-NoAug})$, one-sided Wilcoxon).

D. Phase B: causal intervention

From bar 9 of a key-stable 8-bar prompt, a sustained edit replaces the key component at one layer: $h \leftarrow h - P_V h + P_V \mu_{\kappa^*}$, where V is an orthonormal key subspace and μ_{κ^*} the class-conditional mean of the donor key. Subspaces compared: **V-PROBE** (probe row space, rank 24), **V-MEAN** (class-mean span), **V-DAS** (interchange-trained, ranks 8/24). Controls: **K1** rank/norm-matched random subspace; **K2** sham edit, required to be bit-identical to the clean run (passed, 100/100 prompts); **K3** the subspace of a different layer; **K4** behavioral reference — transposing the prompt itself. Continuations (16 bars, 12 major targets $\times$ 100 prompts $\times$ 8 layers, paired sampling noise) are scored by KS key estimation (TKR, strict and closely-related-tolerant), IKR, and a grammar guard. The musicality

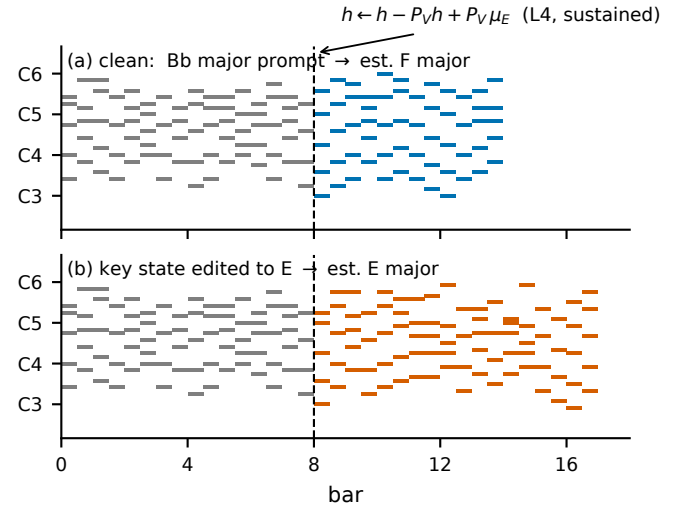


Fig. 1. The claim in one example (real model output, deterministic seed). The same 8-bar Bb-major prompt, continued (a) clean and (b) with the key component of layer-4 activations replaced by the E-major class mean from bar 9 onward. Nothing in the input changes; the continuation’s estimated key follows the injected internal state.

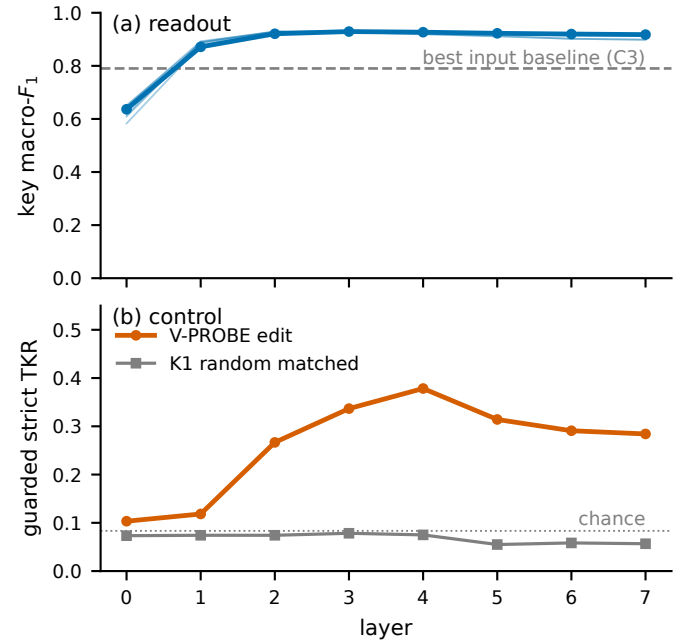


Fig. 2. The key is read and used in the same middle layers. (a) Linear-probe macro- F_1 per layer for all six M-CTRL models (thin lines; R-Aug seed 0 highlighted) against the best input baseline. (b) Causal effect [s0]: guarded strict TKR of the V-PROBE edit vs. the K1 matched random control.

budget $\delta_{\text{PPL}}=0.613$ (90th percentile of the M-REF perplexity rise across 7,989 natural modulations in validation data) was frozen and ledgered *before* any edit ran; edits exceeding it do not count as successes. DR-H3 requires $\text{TKR}(\text{edit}) > \text{TKR}(\text{K1})$ with Holm-corrected Wilcoxon $p < .05$ in $\geq 8/12$ targets at the best in-budget condition.

TABLE II
KEY DECODING AT THE PEAK LAYER (L4, R-AUG SEED 0), 24-CLASS
MACRO- F_1 ON HELD-OUT SEQUENCES. DR-H1 SUPPORTED IN 6/6
MODELS.

linear probe on $h_4(t)$	0.927
MLP probe (secondary)	0.932
best input baseline (C3: PC-hist LR, $W=64$)	0.790
untrained model (C2)	0.243
selectivity control task (C1b)	0.040
corrected margin, BCa 95% CI	+0.105 [0.091, 0.114]

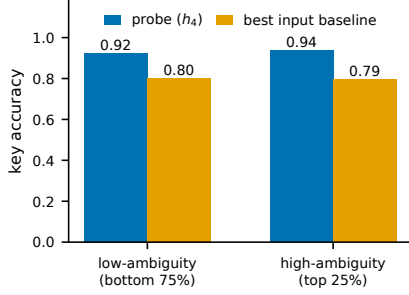


Fig. 3. Probe vs. input baseline by ambiguity stratum (L4, [s0] model): the probe’s margin is largest where surface statistics are least informative.

III. RESULTS

Unless noted, intervention numbers are from the R-Aug seed-0 sweep [s0]; probing results hold for all six M-CTRL models, and the seed-1 sweep replicates every Phase-B verdict (Sec. III-E).

A. H1 — the key is decodable beyond the input surface

Table II and Fig. 2a: the corrected probe–baseline margin excludes zero in layers 1–7 of every model (peak +0.105 at L4), and only fails at L0 (−0.20) — the key is *computed*, not transcribed from the input. In the top ambiguity quartile (KS posterior entropy), probe accuracy is 0.936 vs. 0.795 for the best baseline (Fig. 3); the internal state is most valuable exactly where the surface is least informative.

B. H2 — fifths geometry; equivariant but not a group

Projected to two principal components, the twelve major-tonic probe directions order themselves as C–G–D–A–E–B–F♯... (Fig. 4): perfect circle-of-fifths neighbor agreement (1.00) vs. none for the chromatic ordering (0.00), at every probed layer (exploratory, no decision rule). Transposition acts as a linear map: Procrustes R_k fit on half the sequences aligns held-out activations with 0.05–0.11 relative error. Yet the family $\{R_k\}$ is far from cyclic (Fig. 5: $\varepsilon_{\text{cyc}} \approx 0.98$ against ≈ 1.41 for random orthogonal maps), and — our pre-registered negative result — transposition augmentation does not improve it (3 seeds per regime: R-Aug 0.983 vs. R-NoAug 0.981, one-sided $p = 0.83$); DR-H2b is *not supported*. The

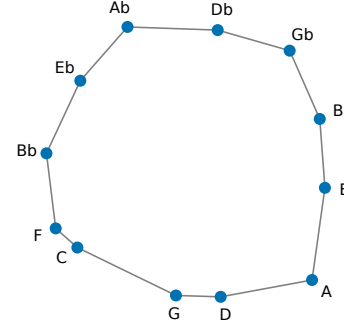


Fig. 4. The twelve major-tonic probe directions (L4, first two principal components) self-organize into a circle of fifths. No key, chord, or degree token exists in the training vocabulary.

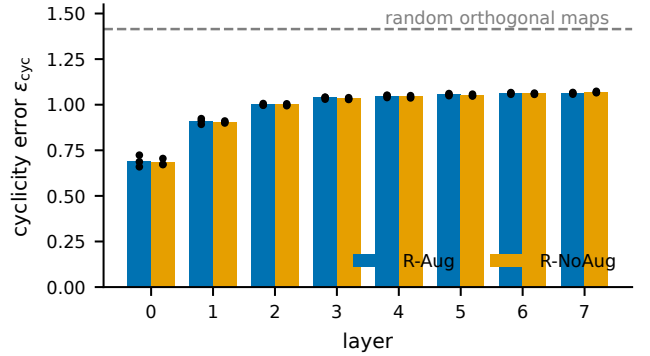


Fig. 5. Cyclicity error of the transposition maps per layer (bars: mean of 3 seeds per regime; dots: individual seeds; dashed: random-orthogonal reference). Transposition augmentation leaves the (non-cyclic) structure unchanged — the pre-registered DR-H2b is not supported.

representation is transposition-equivariant in effect but not organized as a cyclic group, and the training-data key diversity appears to suffice.

C. H3 — editing the key state changes the composed key

Table III and Fig. 6: at the best in-budget condition (V-PROBE, L4), $\text{TKR}(\text{edit}) - \text{TKR}(\text{K1}) > 0$ holds for **12/12** targets (Holm-corrected $p < 10^{-4}$ each, rank-biserial $r = 0.64$ –1.00): DR-H3 is **supported**, with $\geq 8/12$ pre-registered as the bar. Pitch content triangulates the same conclusion without the key estimator: $\text{IKR}_{\kappa^*} = 0.937$ vs. $\text{IKR}_{\kappa_{\text{src}}} = 0.646$ after editing. The specificity matrix (Fig. 7) shows continuations concentrating on the injected key with the estimator’s known fifths-neighbor confusions beside the diagonal and a faint relative-minor band; success decreases gently with target distance on the circle of fifths (Fig. 8; 0.395 at distance 1 to 0.280 at the tritone), while random edits collapse to zero once the identity target is excluded. The edit reaches

58% of the behavioral upper bound (K4), quantifying an *actionability gap*: the state is readable in a rank-24 subspace of one layer, but generation draws on redundant,

TABLE III
GUARDED STRICT TKR (CONTINUATION LANDS EXACTLY ON THE
INJECTED KEY AND STAYS WITHIN THE FROZEN MUSICALITY BUDGET)
[s0]. CHANCE = $1/12 \approx 0.083$.

condition	best layer	guarded stric
V-PROBE edit	L4	0.378 (tolera
V-MEAN edit	L2	0.346
V-DAS edit ($r=24$)	L4	0.213
V-DAS edit ($r=8$)	—	≤ 0.10
K1 random matched	(pooled)	0.068
K3 layer-shuffled	(pooled)	0.160
K4 transposed prompt (upper ref.)	—	0.652 (tolera

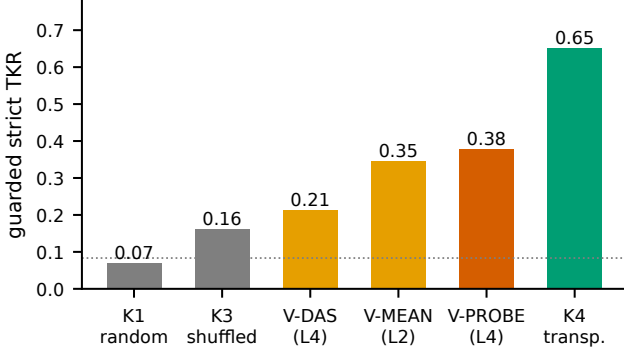


Fig. 6. Guarded strict TKR across conditions [s0] (dotted line: chance $1/12$): the probe-subspace edit at its best layer moves the composed key at $5.6\times$ the matched random control and reaches 58% of the transposed-prompt behavioral reference.

distributed copies that a single-layer edit only partially overrides. Notably, the *optimized* intervention subspace (V-DAS) underperforms the probe’s readout directions, and V-MEAN edits destroy output quality at deep layers (guard pass $\leq 5\%$ at L4–L5) — the guard is load-bearing.

D. H5 — causal efficacy localizes to middle layers

The layer profile of the edit-minus-control effect is single-peaked: 0.03, 0.04, 0.19, 0.26, **0.30**, 0.26, 0.23, 0.23 for L0–L7, peaking at the same layer where probe F_1 peaks; best-vs-median bootstrap CI [0.038, 0.109] excludes zero (DR-H5 supported). Where the key is read is where it acts.

E. Replication and capacity

Second seed. The independently trained R-Aug seed-1 model reproduces every Phase-B verdict: DR-H3 supported with V-PROBE significant in 12/12 targets (TKR 0.23–0.41 vs. K1 0.02–0.10), and DR-H5 supported (CI [0.122, 0.181]). Its single-peaked layer profile peaks at L2 rather than L4, so the *localization* replicates while its exact depth does not; we report both profiles rather than a pooled one.

Capacity (Fig. 9). Training the same data and protocol at 1.6M (L2, $d=128$), 6M (L4, $d=256$), 25M (main line) and 85M (L12, $d=768$) parameters, next-token top-1 is *flat*

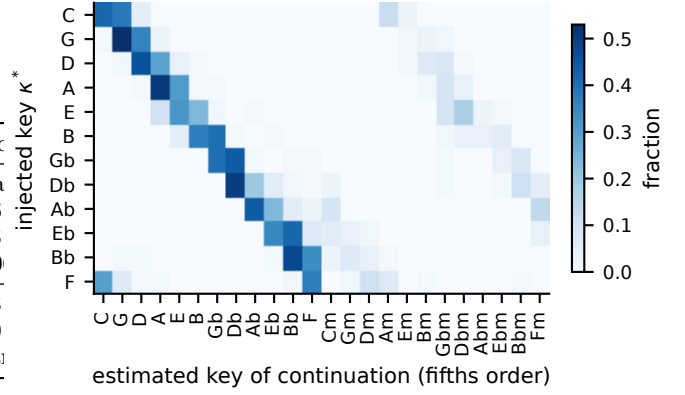


Fig. 7. Specificity [s0]: distribution of the continuation’s estimated key (columns, fifths order) per injected key (rows), V-PROBE edit at L4.

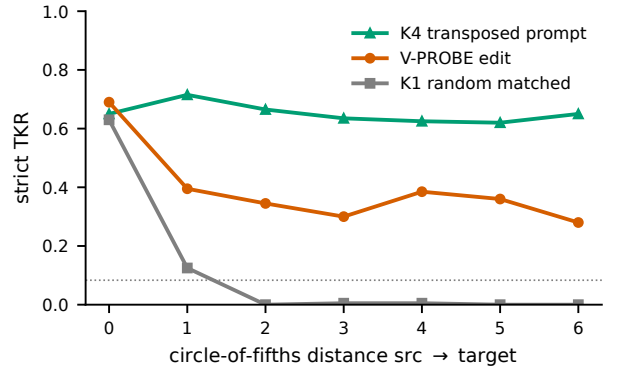


Fig. 8. Strict TKR vs. circle-of-fifths distance from the prompt’s key to the injected key [s0]: V-PROBE edit against the K1 control and the K4 behavioral reference. At distance 0 (identity target) all conditions trivially preserve the key.

across a $50\times$ capacity range (0.868–0.879), yet the internal key representation is not: at 1.6M the probe (peak F_1 0.63–0.66) falls *below* the input baseline (corrected margin $-0.21/-0.17$; DR-H1 *not supported* in either seed), while from 6M upward it clears the baseline and saturates by 25M. A model that is behaviorally equivalent on next-token prediction can therefore lack the world model entirely — it is the surface-statistics account made flesh, and our probe correctly refuses to credit it with a key state. Whether the causal effect emerges at the same threshold: **[TBD (run required): size-model intervention sweeps running]**. (The sweep varies depth and width together; disentangling them is left to future work.)

IV. DISCUSSION AND LIMITATIONS

Together the three levels support the world-model interpretation: a Transformer trained on notes alone maintains a mid-layer key variable with music-theoretic geometry, and generation consults it. The gap to the behavioral bound, the failure of single-layer edits at L0–L1, and the partial effect of layer-shuffled subspaces (K3 = 0.160, between K1 and the true edit) all point to a

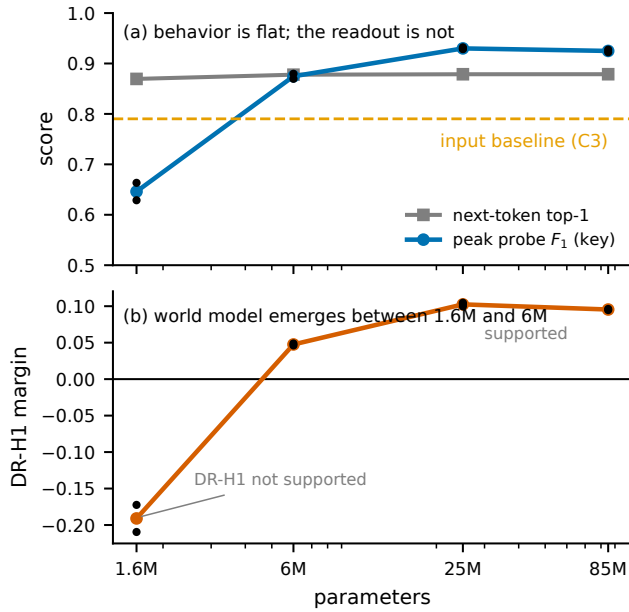


Fig. 9. Behavioral equivalence, representational dissociation. Across a $50\times$ capacity range next-token accuracy is flat, but the key readout is not: below $\sim 6M$ parameters the probe cannot beat the input baseline and the pre-registered DR-H1 margin is negative. A behaviorally equivalent model can lack the world model entirely. Dots: seeds.

distributed, redundantly re-estimated state rather than a single canonical register — a contrast with Othello-GPT’s near-complete editability that we attribute to music’s continual re-grounding of tonality in the immediately preceding notes.

Limitations. M-CTRL is small and trained on synthetic music; claims are about emergence under this training condition, and the capacity sweep varies depth and width together. Remaining robustness additions: probe validation on real annotated music and a probe of a public large symbolic checkpoint. The KS estimator confuses fifths-neighbors, which strict/tolerant reporting brackets. Whether edited continuations *modulate musically* (pivot-like transitions vs. jumps) is deferred to the journal version.

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